

Machine Learning

Final Project - 2026

Objectives:

The goal of this final project is to apply deep learning techniques to a multimodal food classification problem. Each student will build a model that predicts the name of a Taiwanese food dish using:

1. a food image, and
2. a text description of the dish ingredients.

Each student will train and evaluate their own model and compete individually on a private Kaggle leaderboard using a hidden test set.

Instruction

Project Instruction

1. Train your own multimodal deep learning classifier using the provided dataset.
2. Your model must use the image modality and the text-description modality. Explain clearly how the two modalities are used or combined.
3. Build and train your model using only the provided dataset. **Do not add external images, extra labels, or pretrained weights.**
4. You may train directly in Kaggle Notebooks, Google Colab, or on your local machine.
5. Your final Kaggle output must be a submission.csv file following the required format (Check the template in Kaggle).
6. Submit your project materials to E3, including codes, the final weight, and final submission.csv.

Dataset Details:

Split	Size	Label Information
Training folder train/	60 images x 25 classes = 1,500 images	Class folder name is the label
Test folder test/	4,832 images, flat folder	Hidden labels; evaluated only by Kaggle

25 Taiwanese Food Classes

Classes				
Beef Noodles	Braised Pork Over Rice	Bubble Tea	Coffin Toast	Deep-Fried Chicken Cutlets
Egg Pancake Roll	Grilled Taiwanese Sausage	Iron Egg	Mango Shaved Ice	Mochi

Oyster Omelet	Papaya Milk	Pineapple Cake	Potsticker	Rice Dumpling
Scallion Pancake	Stinky Tofu	Sweet Potato Ball	Tanghulu	Taro Ball
Three-Cup Chicken	Turkey Rice	Turnip Cake	Xiaolongbao	Yolk Pastry

Extra Files Provided

File	Description
class_labels.csv	Maps each class name to a numeric label ID from 0 to 24
taiwanese_food_descriptions.csv	Full ingredient list and text description for each of the 25 classes used as the text modality during training
test_recipes.csv	Partial ingredient hint (2–3 random ingredients) for every test image used as the text modality during inference
sample_submission.csv	Template showing the required format for your submission file

You may divide the training set into your own train/validation split as needed.

Task Overview

This is a multimodal classification problem — your model must use two sources of information at both training and test time:

1. Image modality — a photo of the food dish
2. Text modality — an ingredient list describing what is in the dish

During training

Each labeled image has a known class, so you automatically know its full ingredient description from `taiwanese_food_descriptions.csv`. For example, a `bubble_tea` image is always paired with:

"black tea, tapioca pearls, whole milk, sugar syrup, ice cubes. Cold sweetened milk tea..."

During inference (test set)

Every test image in `test_recipes.csv` comes with a partial ingredient hint, only 2 or 3 ingredients are shown, chosen randomly. For example:

```
img_00041.jpg, tapioca pearls, whole milk
img_00123.jpg, rice flour, shallot
img_00456.jpg, chicken pieces, fresh thai basil
```

The hint alone is not enough to identify the food, "rice flour, shallot" could be turnip_cake, scallion_pancake, or oyster_omelet. Your model must look at both the image and the hint together to make the correct prediction.

The 50/50 rule

When combining image and text scores at inference time, you must use equal weight:

$$\text{final_score} = 0.5 \times \text{image_score} + 0.5 \times \text{text_score}$$

Before combining, convert both image logits and text logits into probability scores using softmax. The 50/50 rule applies to the final prediction score during inference/submission. During training, you may design your own fusion architecture.

A model that assigns more than 50% weight to the image (or ignores the text entirely) does not satisfy the multimodal requirement.

Starter Code

One template is provided to help you get started:

File	Purpose
student_train_scratch.py	Training template, fill in the TODOs to build your model, train it, and generate submission.csv

The template gives you the dataset loading, submission generation, and training loop skeleton. You must design and implement your own image encoder and text encoder from scratch, **no downloaded pretrained weights allowed**.

student_train_scratch.py does everything in one run:

1. Trains your multimodal model on the 1,500 labeled images
2. Saves the best weights to my_model_scratch.pth
3. Loads test_recipes.csv and runs inference on all 4,832 test images
4. Saves submission.csv, ready to upload to Kaggle

What You Can Do

Allowed	Notes
Design any image encoder architecture	CNN, ResNet-style, as long as you build and train it yourself

Design any text encoder architecture	RNN, LSTM, GRU, built and trained by you
Design your own internal fusion strategy during training	Cross-attention, concatenation, weighted sum
Use data augmentation	Random crop, flip, color jitter, etc.
Create your own train/val split	From the 1,500 training images
Tune hyperparameters	Learning rate, batch size, epochs, etc.
Train on Kaggle Notebooks, locally, or on Google Colab	Free GPU available on Kaggle

What You **Cannot** Do

Not Allowed	Notes
Use any pretrained model (CLIP, ResNet, ViT, BERT, etc.)	Your image encoder and text encoder must be built and trained entirely by you from the provided dataset only
Use only image or only text – your model must be multimodal	Using a single modality is not the task
Assign more than 50% weight to image or more than 50% weight to text at inference	The 50/50 rule is required — final score must be $0.5 \times \text{image_score} + 0.5 \times \text{text_score}$
Ignore test_recipes.csv when generating submission.csv	Every test image has a recipe hint; it must be used
Use external food datasets or labels not provided	Unfair advantage and violates data rules

Kaggle Competition Registration:

- To join the final project, use this link:
<https://www.kaggle.com/t/bf9ccb1d13d64acfb85c5c2fe82b9d94>
- Your Kaggle account must use the same email as your E3 account. If you used a different email or can not join, email the TAs immediately.
- Only one Kaggle account per student is allowed.
- After joining the competition, change your Kaggle display name to: StudentID_Name (e.g., 411540006_桑佳亞)
- ⚠ Failing to do so means you won't be able to submit your project.

Rules:

1. Individual Work Only: This is an individual project. No teamwork, code sharing, or joint submissions allowed.
2. Data Usage: Training images and class labels are provided. Test images are provided, but labels are hidden. Evaluation is done via Kaggle submissions only.
3. Submission Limits: Max 20 submissions per day on Kaggle.
4. Evaluation Metric: The competition is evaluated using classification accuracy. A higher score means better classification performance.
5. Submission deadline: Final submission must be made on Kaggle before June 1, 10:00am
6. Baseline: Your model must achieve at least the baseline score **0.5**. Students who fail to beat the baseline model will receive a significantly lower score.
7. Code review during submission will be used to verify compliance.

Scoring System:

1. Top 15 on Leaderboard → Finalist Round
 - a. Final Score = 50% Kaggle Rank (85–99) + 50% In-Class Presentation (85–99)
 - b. Presentation scores will be peer-reviewed via Google Form during class.
 - c. Students ranked 1st to 15th on the leaderboard must:
 - i. Prepare a presentation and submit it on E3 before In-Class Presentation
 - ii. Their final score will be recalculated and re-ranked based on the average of Kaggle and presentation scores.
 - iii. If multiple students have the same final score, they will receive the same grade, and the next rank will skip accordingly (e.g., 99, 99, 99, then 96).
2. Below Top 15
 - a. Final score will be ≤ 84 , depending on your Kaggle rank and performance.
 - b. Students who fail to beat the baseline model will receive a score below 40.

